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GEN AI ARCHITECT JOURNEY · PHASE-004

Basic FastAPI App

Project brief and build specification

Goal	Build a very basic web API using FastAPI
Difficulty	Beginner
Tools	Python · FastAPI · Uvicorn
Single file	main.py
Phase	Phase-004 - Interfaces and APIs
Folder	Phase-004-FastAPI-And-Streamlit

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1. What you will build

A very basic web API using the FastAPI framework.

FastAPI lets you create web endpoints in Python that return data - usually JSON - when someone visits a URL. Your task is to set up a clean Python environment, install FastAPI, write a small app in a **.py** file, run it with a server, and view it working in the browser. FastAPI also generates interactive API documentation for you automatically.

Difficulty	Beginner
Tools	Python, FastAPI, Uvicorn

2. What your app must include

- Use the **FastAPI** framework to build the app.
- Create a root endpoint **GET /** that returns a simple JSON response - for example, a welcome message.
- Add at least one more endpoint that takes a value from the URL - either a path parameter such as **/greet/{name}** or a query parameter - and returns it inside the response.
- Run the app with the **Uvicorn** server and confirm it works in the browser.
- Open the automatic interactive docs at **/docs** and check your endpoints there.
- Keep all your code in a single file named **main.py**.

Think it through

FastAPI builds the interactive docs from your code, so clear endpoint names and return values matter. Think about what each endpoint should return, what happens when a parameter is missing, and how the JSON response should be shaped. Keep it simple - this is meant to be a first, small API.

3. Step-by-step setup

Follow these steps in order. Commands are shown for both macOS/Linux and Windows where they differ.

Step 1 - Create a project folder

```
SHELL
mkdir fastapi-basic-app
cd fastapi-basic-app
```

Step 2 - Create a virtual environment (venv)

A virtual environment keeps this project isolated from the rest of your system.

```
SHELL
python -m venv venv
# (on some systems use 'python3' instead of 'python')
```



Step 3 - Activate the virtual environment

```
SHELL
# macOS / Linux
source venv/bin/activate

# Windows (PowerShell)
venv\Scripts\Activate.ps1

# Windows (Command Prompt)
venv\Scripts\activate.bat
```

When activated, you will see **(venv)** at the start of your terminal prompt.

Step 4 - Install FastAPI and the server

Install FastAPI together with Uvicorn, the server that runs your app.

```
SHELL
pip install fastapi uvicorn
```

Step 5 - Run your app

Start the app with Uvicorn. Replace **file_name** with the name of your Python file and **app** with your FastAPI variable name.

```
SHELL
uvicorn file_name:app --reload
```

The app runs at **http://127.0.0.1:8000**, and the interactive docs are at **http://127.0.0.1:8000/docs**.

Step 6 - Stop and deactivate when finished

Press **Ctrl + C** in the terminal to stop the server, then deactivate the venv.

```
SHELL
deactivate
```

4. How it should behave

Your wording can differ, but visiting the root endpoint should return JSON similar to this:

```
HTTP
GET http://127.0.0.1:8000/

{
  "message": "Hello, FastAPI"
}
```



5. Deliverables

- All application code in a single file named **main.py**.
- A **requirements.txt** so the project can be recreated on any machine.
- A **README.md** documenting every endpoint with request and response examples.
- An importable **Postman collection** that proves each endpoint behaves as documented.
- The whole project committed to GitHub.

Definition of done

The server starts cleanly, every documented endpoint returns what the README says it returns, invalid input is rejected with the correct status code, and the interactive docs at **/docs** render every route.

Next

The completed build for this brief is in this folder: see **main.py** and **README.md**.

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